

SOFTWARE ENGINEERING LAB - 1

Lab 1: Requirements Engineering & UML Use-Case Modelling

Team Number: 08

Team Members:

Krrish Singla (PES1UG24AM141)
Kishan K Shenoy (PES1UG24AM138)
Kushal G (PES1UG24AM145)
Namith Kumar C (PES1UG24AM171)

Problem Statement #08 — Digital Campus Library Reservation Gateway

Target Stakeholders / Actors: Student Member, Head Librarian

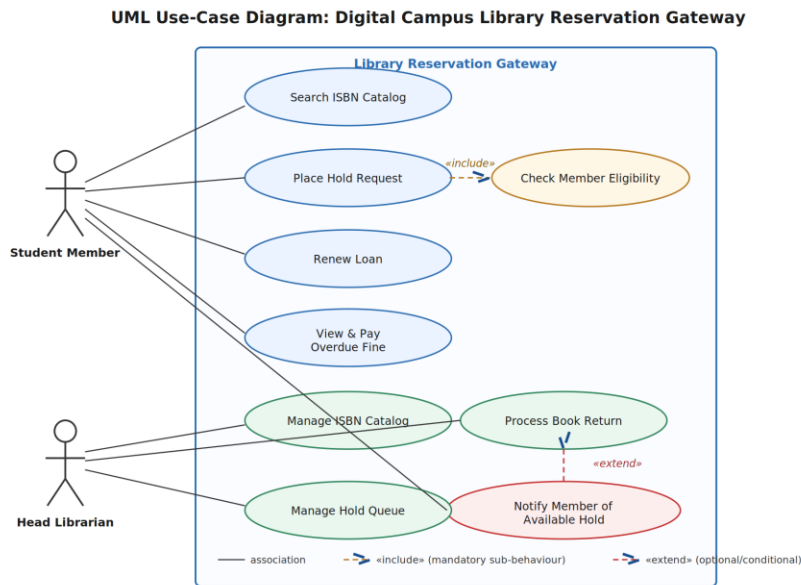
1. Requirements Table

Five Functional Requirements (FR-001–FR-005, with FR-001 as given) and two Non-functional Requirements (NFR-001–NFR-002, with NFR-001 as given).

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow members to search the ISBN repository and place hold requests on currently loaned books to enter a FIFO waitlist.	High	Pass: User is assigned correct position in queue. Fail: User already holding max allowed books places an additional reservation.	Enables equitable, order-based access to high-demand titles without manual librarian intervention.
FR-002	Functional	The system shall automatically notify a member (email/SMS) within 5 minutes of a held book being checked in, and hold the copy for 48 hours before releasing it to the next member in the queue.	High	Pass: Notification sent within 5 minutes and copy auto-released after 48 hours if uncollected. Fail: Notification not sent or copy held indefinitely.	Prevents held copies from sitting idle and ensures timely, fair circulation of returned titles.
FR-003	Functional	The system shall automatically calculate and post an overdue-fine transaction to a member's account for each day a borrowed book remains unreturned past its due date.	High	Pass: Fine amount equals (days overdue x daily rate) and appears in transaction history by end of due date. Fail: No fine posted or incorrect amount calculated.	Automates fine accrual, reducing manual librarian effort and encouraging timely returns.
FR-004	Functional	The system shall allow the Head Librarian to add, update, or deactivate ISBN catalog records, including title metadata and physical copy counts.	Medium	Pass: Catalog change is reflected in member search results within 1 minute of saving. Fail: Change is not persisted or not visible to members.	Keeps the searchable catalog accurate as the physical collection changes over time.
FR-005	Functional	The system shall allow a member to renew a currently borrowed book for one additional loan period, provided no other member holds an active reservation on that title.	Medium	Pass: Due date is extended by the standard loan period and confirmation is shown. Fail: Renewal is blocked when an active hold exists on the title.	Gives members flexibility to keep books longer while protecting the FIFO queue's fairness for waiting members.
NFR-001	Non-functional	The catalog search query shall return paginated results for 50,000 titles in less than 150 ms.	High	Pass: Benchmarking tests confirm target latency and security standards are met under simulated peak load.	Ensures a responsive search experience during peak enrollment/exam periods when catalog usage spikes.
NFR-002	Non-functional	The system shall be available 99.9% of the time during the academic semester (excluding scheduled maintenance) and shall automatically recover from a single-node failure within 5 minutes.	High	Pass: Uptime monitoring confirms $\geq 99.9\%$ availability over a rolling 30-day window and failover tests recover service within 5 minutes. Fail: Downtime exceeds threshold or requires manual recovery.	Hold placement, renewal, and fine payment are time-sensitive actions members rely on throughout the day.

2. UML Use-Case Diagram

Actors: Student Member and Head Librarian. The diagram below shows all primary use cases along with the modelled «include» and «extend» relationships.



Include relationship: "Place Hold Request" includes "Check Member Eligibility" (eligibility is always verified before a hold is created).

Extend relationship: "Notify Member of Available Hold" extends "Process Book Return" (only triggered when the returned title has an active hold queue).

3. Use-Case Flow Specification

Use Case Name	View & Pay Overdue Fine
Related Requirement	FR-003
Primary Actor	Student Member
Preconditions	1) The Student Member is authenticated and logged in. 2) At least one overdue fine transaction exists on the member's account.
Postconditions	Success: The fine balance is reduced by the payment amount and the payment is recorded; if the balance reaches zero, borrowing privileges are restored. Failure: The fine balance remains unchanged.

Main Success Scenario

1. Student Member navigates to the "My Fines" section of their dashboard.
2. System displays the list of outstanding fine transactions with amounts and dates incurred.
3. Student selects "Pay Now" and enters payment details.
4. System validates the payment method.
5. System processes the payment through the payment gateway.
6. System updates the fine ledger, marking the transaction as paid.
7. If the account balance reaches zero, the system restores the member's borrowing privileges.
8. System displays a payment confirmation and the updated balance.

Alternate Flow A1: Payment Method Declined

- A1.1. At step 4 or 5, the payment gateway declines the transaction.
- A1.2. System displays: "Payment could not be processed. Please try another method."
- A1.3. The fine balance remains unchanged and borrowing privileges, if suspended, remain suspended.
- A1.4. Use case ends; the student may retry with a different payment method.